

For questions 1-6, factor each expression by grouping.

1. $2y^4 - 6y^3 + 14y^2 - 42y$

$$\begin{aligned} & y(2y^3 - 6y^2 + 14y - 42) \\ & y[(2y^3 - 6y^2) + (14y - 42)] \\ & y[2y^2(y - 3) + 14(y - 3)] \\ & y(2y^2 + 14)(y - 3) \\ & 2y(y^2 + 7)(y - 3) \end{aligned}$$

2. $15x^2 + 34x + 15$

$$\begin{aligned} & (15x^2 + 9x) + (25x + 15) \\ & 3x(5x + 3) + 5(5x + 3) \\ & (3x + 5)(5x + 3) \end{aligned}$$

3. $-6a + 24ab - 20b + 5$

$$\begin{aligned} & (-6a + 24ab) + (-20b + 5) \\ & -6a(1 - 4b) + 5(-4b + 1) \\ & (-6a + 5)(-4b + 1) \end{aligned}$$

4. $6x^2 - x - 40$

$$\begin{aligned} & (6x^2 - 16x) + (15x - 40) \\ & 2x(3x - 8) + 5(3x - 8) \\ & (2x + 5)(3x - 8) \end{aligned}$$

5. $3x^2 + 5x + 6x + 10$

$$\begin{aligned} & (3x^2 + 5x) + (6x + 10) \\ & x(3x + 5) + 2(3x + 5) \\ & (x + 2)(3x + 5) \end{aligned}$$

6. $5x^2 + 52x + 63$

$$\begin{aligned} & (5x^2 + 45x) + (7x + 63) \\ & 5x(x + 9) + 7(x + 9) \\ & (5x + 7)(x + 9) \end{aligned}$$

For questions 7-10, consider the trinomial $16x^2 + 16x - 5$.

7. List all the factor pairs for -80 .

$-80, 1$ $-40, 2$ $-20, 4$ $-16, 5$ $-10, 8$
 $80, -1$ $40, -2$ $20, -4$ $16, -5$ $10, -8$

8. Find the factor pair for -80 that add to the sum of 16.

$20, -4$

9. Use question 8 to write the expanded form of $16x^2 + 16x - 5$ that can be used to factor the expression by grouping.

$(16x^2 - 4x) + (20x - 5)$

10. Factor the expression $16x^2 + 16x - 5$ by grouping.

$4x(4x - 1) + 5(4x - 1)$
 $(4x + 5)(4x - 1)$